

Ultimate Nutrition

Premium MCT Gold

100% medium chain triglycerides oil

Many people think that well developed muscles are evidence of physical fitness. Firm, toned muscles are considered esthetically pleasing and hundreds of thousands of people spend untold hours to appear "muscular." Their appearance notwithstanding, muscles are specialized tissues that contract to produce body movement.

Muscle cells require great energy to contract, especially during intense exercise and endurance training. Nature has evolved a sophisticated system to produce energy that the body needs. Energy is released when muscle cells break down carbohydrate, fat and protein. The body utilizes these nutrients in rather specific amounts during physical activity. Carbohydrates are the body's most readily available source of nutrient energy. Carbohydrates are digested and metabolized to generate glucose, which serves as the immediate source of fuel. Excess glucose that is not used directly to yield energy is transported via the bloodstream to liver and muscle tissue, where it is stored in the form of glycogen.

In contrast, fat is the body's most rich source of food energy. Unlike glycogen, fat is the stored in the fat cells and, therefore, is less readily available for energy production. In order for fat to be used as an energy source, it must first be broken down into fatty acids, which are then transported by the blood to the muscle to be burned. Protein is required for the repair and recovery of the muscle tissue.

Intense exercise on a regular basis can induce muscle fatigue. Thus, people are recognizing more and more that carbohydrates are an essential factor in preventing muscle fatigue. Carbohydrates, especially refined sugars, however, are not ideal source of immediate energy needed to maintain an exercise regimen. That is where Medium Chain Triglycerides (MCTs) can help the athlete develop stamina and endurance.

Medium chain triglycerides are actually not carbohydrates. Rather, MCTs are important fatty acids that provide energy exactly like carbohydrates. Also known as caprylic and capric acids, MCTs function differently from conventional fats. That means that less fat is absorbed in the body. In intense exercise, MCTs prevent the breakdown of the muscle tissue, since they produce the so-called ketones, which are used directly by the muscle to produce energy and reduce muscle loss. Conventional fats do not produce many ketones. In addition, the MCTs are quickly absorbed and circulate in the bloodstream, and they do not produce fatigue as is common with the consumption of simple sugars.

The MCTs are unique in that they can be turned into energy inside the mitochondria --the powerhouses of energy production in the cell-- in the presence of carbohydrates. Conventional fats can be burned only after carbohydrate reservoir of the cell has been

depleted. This has important implication in that the burning of MCTs in the presence of carbohydrates spares glycogen and, hence, production of ketones and sparing of glycogen both prolong endurance for training and stamina.

Medium chain triglycerides also improve the absorption of amino acids, which are critical for muscle tissue repair. Furthermore, MCTS also improve the absorption of calcium and magnesium. These minerals are needed for carbohydrate and amino acid metabolism to improve muscle contraction response time.

The MCTs have been shown to lower cholesterol, and its deposition in various body tissues. In fact, MCTs are digested immediately after ingestion as they are hydrolyzed by the enzyme lipase present in the saliva. Additionally, energy released by the digestion and metabolism of MCTs is converted in the body heat by a process called thermogenesis, which favorably affects the basal rate of metabolism. When the body converts energy into heat, the metabolism increases that, in turn, leads to fat loss.

In short, MCTs are known as "carbo-lipid" for good reason, since they not only prevent muscle loss and increase lean body mass but also lower cholesterol deposition. The MCTs, therefore are a nutrient that has great value for those in endurance training and the "body culture."